

Short Analysis Pipeline

BIOLM0051

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Scenario

A suitcase containing deez several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

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Introduction

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Methods

The FASTQ files in this report are small and the use of HPC resources in this case is not necessary. However, this workflow is designed to scale easily to similar analyses with much larger datasets. All steps here are able to run on an HPC cluster without modification, demonstrating reproducibility, automation, and capacity for high-throughput analyses.

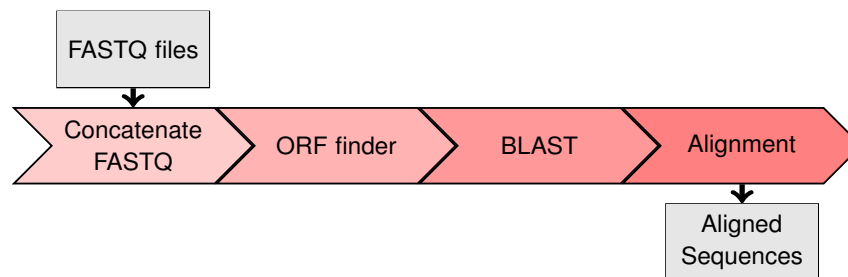


Figure 1: Workflow of Snakemake analysis pipeline.

Some python code:

```
from Bio import SeqIO
for record in SeqIO.parse("sequence.fasta", "fasta"):
    print(record.id, record.seq.translate())
```

Some Bash script:

```
#!/bin/bash
#SBATCH --job-name=blast_job
#SBATCH --partition=teach_cpu
#SBATCH --time=01:00:00
#SBATCH --mem=2000M

blastn -db nt -query sequence.fa -out results.out -outfmt 7 -remote
```

Results

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis

Table 1: Top 15 BLASTP hits for sample OMRGCv2f.11922010.1 against SwissProt.

Query	Subject	% Identity	Alignment Length	E-value	Bitscore
OMRGCv2f.11922010.1	A5UYW4.1	55.56	63	1.3e-17	85.1
OMRGCv2f.11922010.1	A7NS65.1	55.56	63	1.8e-17	83.5

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```
import pandas as pd
data = pd.read_csv("expression.csv")
normalized = data.apply(lambda x: np.log2(x+1))
```

Alignment results using Clustal Omega on some test data:

```
test1  . MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANK 54
test2  . MGDVEKGKKIFVQKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAAGFSYTDANK 54
test3  KAGSAKKGATLKFKTRCLQCHTVEKGGPHKVGPNLHGIFGRHSQAEGYSYTDANI 59

test1  NKGIIWGEDTLMEYLENPKKYIPGTMIFVGIKKKEERADLIAYLKKATNE 105
test2  NKGITWGEDTLMEYLENPKKYIPGTMIFAGIKKKG ERADLIAYLKKATNE 105
test3  KKNVLWDENNMSEYLTNPKKYIPGTMAFGGLKKEKDRNDLITYLK..... 105

[X] non-conserved
[X] ≥ 50% conserved
[X] ≥ 80% conserved
```

Figure 2: Alignment using Clustal Omega and Identity mode in texshade

Discussion

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui.

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References

- [1] WJ Kent. Blat—the blast-like alignment tool. *Genome research*, 12(4):656–664, 2002.
- [2] MJ Wolin, TL Miller, and CS Stewart. Microbe-microbe interactions. In *The rumen microbial ecosystem*, pages 467–491. Springer, 1997.
- [3] B Baker, P Zambryski, B Staskawicz, and SP Dinesh-Kumar. Signaling in plant-microbe interactions. *Science*, 276(5313):726–733, 1997.

Acknowledgments

This work was carried out using the computational facilities of the Advanced Computing Research Centre, University of Bristol - <http://www.bristol.ac.uk/acrc/>

Supplementary Data

Supplementary data can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git